

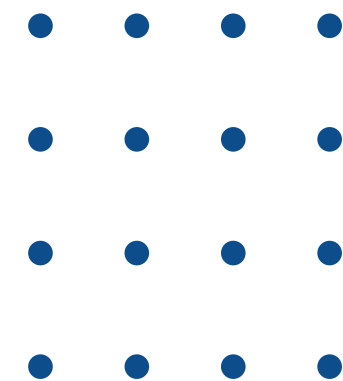
Juillet, 2026



# KNOWLEDGE LEARNING

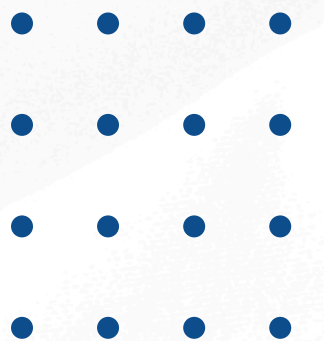
Presented by : ESSOUFI RIM

Centre Européen de Formation



# L'entreprise Knowledge

- Éditeur de livres de formation
- Domaines : Musique, Informatique, Jardinage, Cuisine
- Objectif : Développer une offre e-learning
- Projet : "Knowledge Learning"



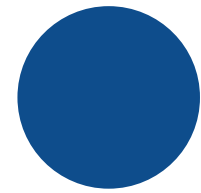
# Contexte et objectifs



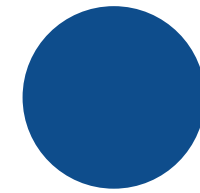
- Créer une plateforme e-learning/e-commerce
- Permettre aux clients d'étudier en autonomie
- Acheter des cursus ou des leçons
- Obtenir des certifications

Technologie : Symfony 8.1

# Fonctionnalités principales



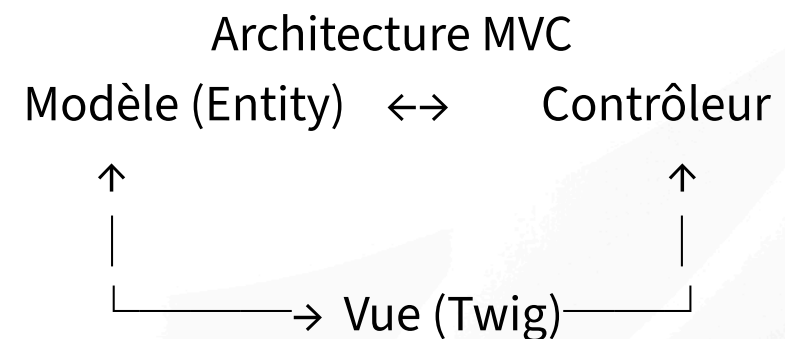
- Création de compte avec activation par email
- Authentification sécurisée (rôles admin/client)
- Achat de cursus et de leçons



- Validation des leçons
- Obtention automatique de certifications
- Backoffice admin

# Architecture MVC

## (Modèle-Vue-Contrôleur)



- Modèle : Doctrine (ORM)

- Vue : Twig

- Contrôleur : PHP

src/Entity/User.php

```
#[ORM\Entity(repositoryClass: UserRepository::class)]
#[UniqueEntity(fields: ['email'], message: 'There is already an account with this
email'')]
class User implements UserInterface, PasswordAuthenticatedUserInterface
{
    #[ORM\Id]
    #[ORM\GeneratedValue]
    #[ORM\Column]
    private ?int $id = null;

    #[ORM\Column(length: 180, unique: true)]
    private ?string $email = null;

    #[ORM\Column]
    private ?string $password = null;
```

# Entité User (Modèle Doctrine)

# Justification de la conception DB



Méthode Merise :

1. Modèle Conceptuel de Données (MCD)
  - Entités : User, Role, Theme, Course, Lesson, Purchase, UserLesson, Certification
  - Relations : 1-N, N-1
2. Modèle Logique de Données (MLD)
  - Tables relationnelles
  - Clés primaires et étrangères
3. Modèle Physique de Données (MPD)
  - Implémentation avec Doctrine

```
#[ORM\Column]
private bool $isActive = false;

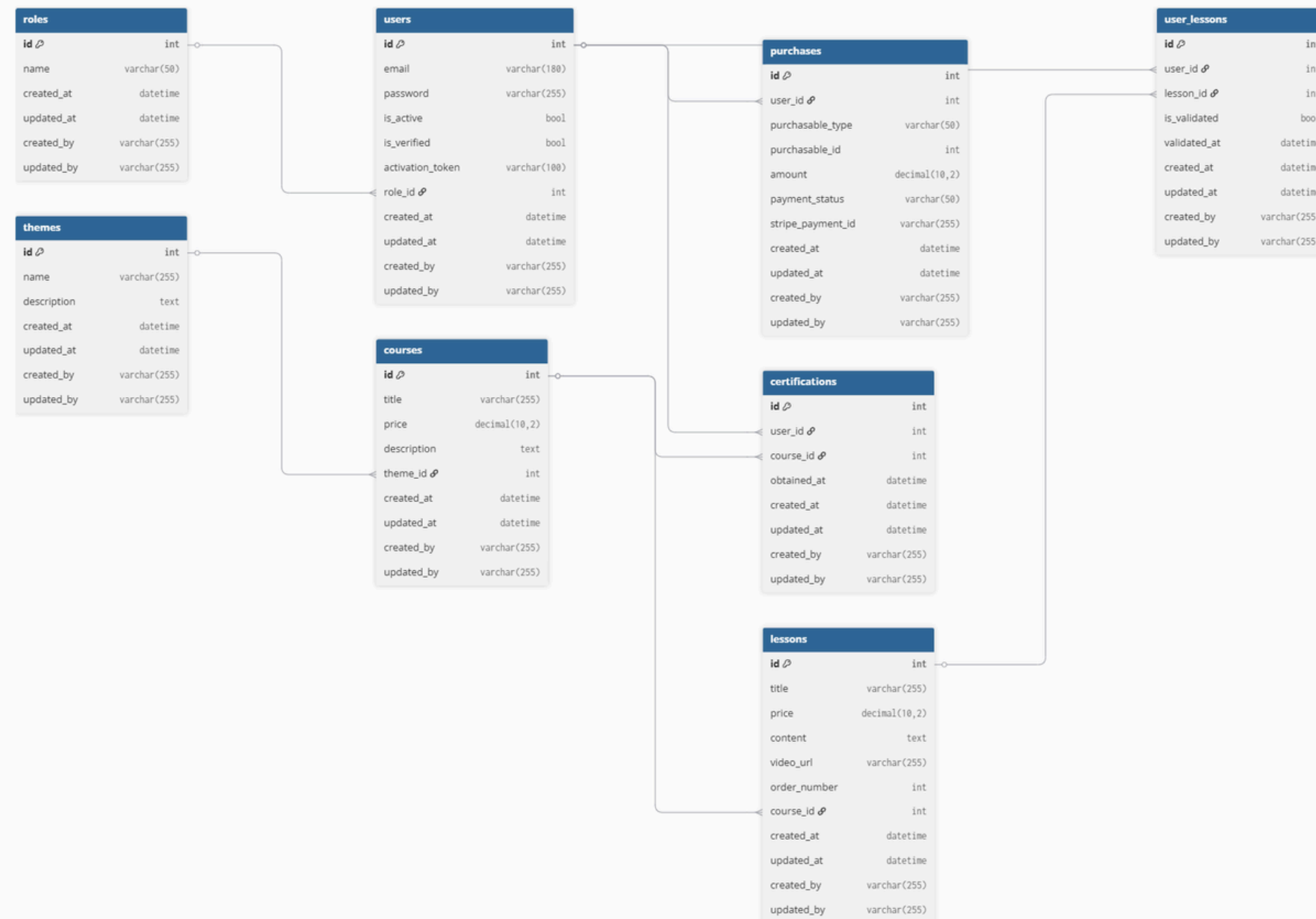
#[ORM\Column]
private bool $isVerified = false;

#[ORM\Column(length: 100, nullable: true)]
private ?string $activationToken = null;

#[ORM\ManyToOne]
#[ORM\JoinColumn(nullable: false)]
private ?Role $role = null;
```

# Sécurité et relations Doctrine

# Schéma de la base de données



## 8 tables principales :

- roles,
- users,
- themes,
- courses,
- lessons,
- purchases,
- user\_lessons,
- certifications

src/Repository/UserRepository.php

```
class UserRepository extends ServiceEntityRepository
{
    public function __construct(ManagerRegistry $registry)
    {
        parent::__construct($registry, User::class);
    }

    public function findActiveUsers(): array
    {
        return $this->createQueryBuilder('u')
            ->andWhere('u.isActive = :active')
            ->setParameter('active', true)
            ->orderBy('u.createdAt', 'DESC')
            ->getQuery()
            ->getResult();
    }
}
```

## Repository personnalisé (ProductRepository)

# Mise en place de la base de données



- Doctrine ORM- Migration : `php bin/console make:migration`
- Création des tables : `php bin/console doctrine:migrations:migrate`
- Fixtures : `php bin/console doctrine:fixtures:load`

Outils : MySQL, Doctrine

# Commandes de migration

```
PS C:\ESERCIZI CEF\knowledge-learning> mkdir migrations
>>

Directory: C:\ESERCIZI CEF\knowledge-learning

Mode                LastWriteTime         Length Name
----                -
d-----         08/10/2026    22:00             migrations

PS C:\ESERCIZI CEF\knowledge-learning> php bin/console make:migration
>>
created: migrations/Version20261008200046.php

Success!

Review the new migration then run it with php bin/console doctrine:migrations:migrate
See https://symfony.com/doc/current/bundles/DoctrineMigrationsBundle/index.html
PS C:\ESERCIZI CEF\knowledge-learning> php bin/console doctrine:migrations:migrate
>>

WARNING! You are about to execute a migration in database "knowledge_learning" that could result in schema changes and data loss. Are you sure you wish to continue? (yes/no) [yes]
> YES

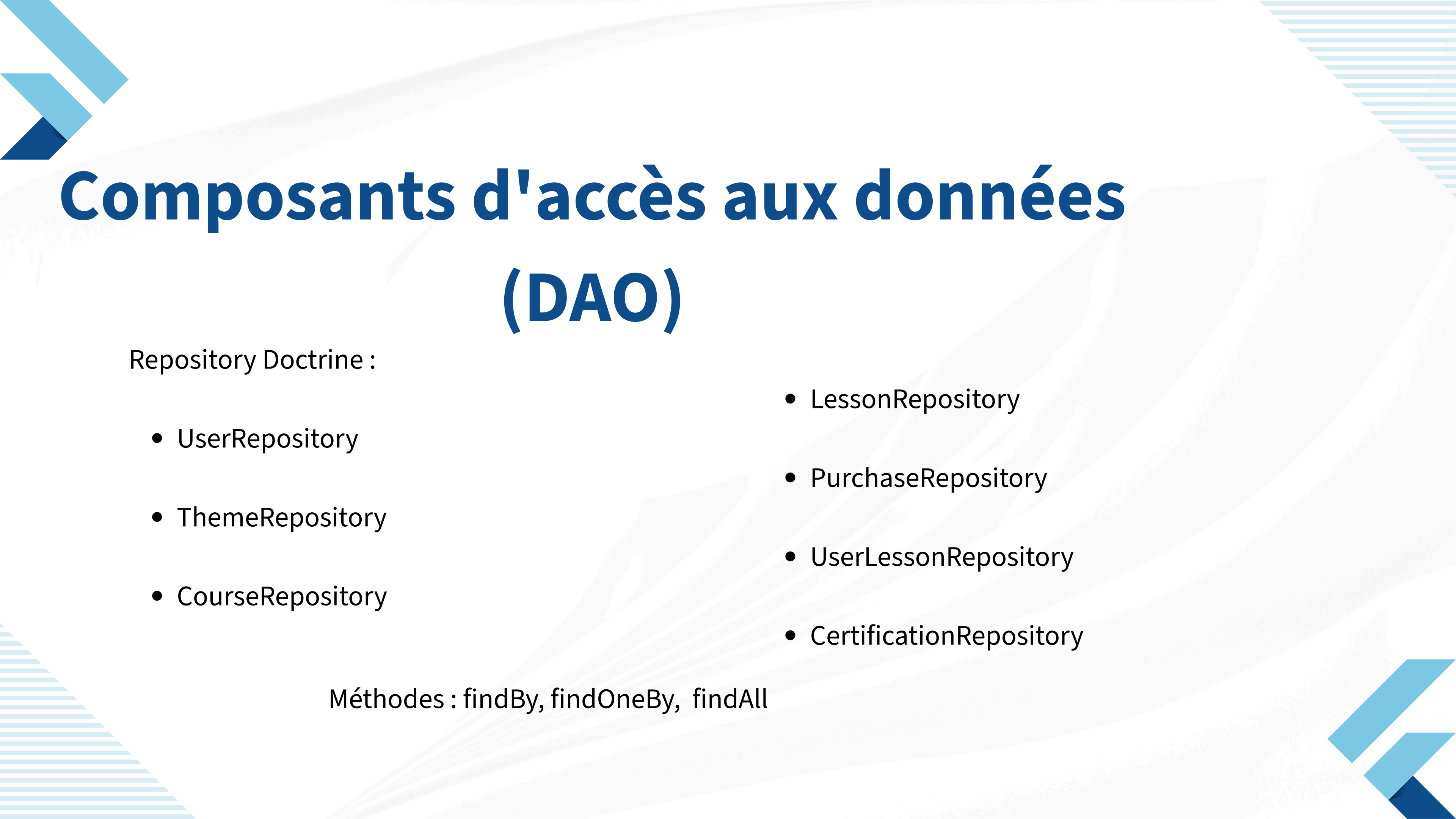
[notice] Migrating up to DoctrineMigrations\Version20261008200046
[notice] finished in 679.1ms, used 16M memory, 1 migrations executed, 8 sql queries

[OK] Successfully migrated to version: DoctrineMigrations\Version20261008200046

>> C:\ESERCIZI CEF\knowledge-learning> php bin/console doctrine:fixtures:load

Careful, database "knowledge_learning" will be purged. Do you want to continue? (yes/no) [no]:
> YES

> purging database
> loading App\DataFixtures\AppFixtures
PS C:\ESERCIZI CEF\knowledge-learning>
```



# Composants d'accès aux données (DAO)

Repository Doctrine :

- UserRepository
- ThemeRepository
- CourseRepository
- LessonRepository
- PurchaseRepository
- UserLessonRepository
- CertificationRepository

Méthodes : findBy, findOneBy, findAll

src/Controller/RegistrationController.php

```
#[Route('/register', name: 'app_register')]

public function register(Request $request, UserPasswordHasherInterface
$userPasswordHasher, EntityManagerInterface $entityManager): Response
{
    $user = new User();

    $form = $this->createForm(RegistrationFormType::class, $user);

    $form->handleRequest($request);

    if ($form->isSubmitted() && $form->isValid()) {
        $user->setPassword(
            $userPasswordHasher->hashPassword(
                $user,
                $form->get('plainPassword')->getData()
            )
        );

        $entityManager->persist($user);

        $entityManager->flush();

        $this->emailVerifier->sendEmailConfirmation('app_verify_email', $user,
            (new TemplatedEmail())
                ->from(new Address('knowledge@learning.com', 'Knowledge Learning'))
                ->to($user->getEmail())
                ->subject('Confirmez votre email')
                ->htmlTemplate('registration/confirmation_email.html.twig')
        );

        return $this->redirectToRoute('app_home');
    }

    return $this->render('registration/register.html.twig', [
        'registrationForm' => $form->createView(),
    ]);
}
```

# Contrôleur d'inscription (RegistrationController)

# Intégration E-commerce (Stripe)

Service : StripeService

- Création de session de paiement
- Mode Sandbox (test)

Routes :

- POST /checkout/{type}/{id}
- GET /checkout/success/{type}/{id}
- -GET /checkout/cancelSécurité : CSRF Protection

src/Service/StripeService.php

```
class StripeService
{
    public function __construct(string $secretKey, string $publishableKey)
    {
        Stripe::setApiKey($secretKey);
    }

    public function createCheckoutSession(array $cart, string $successUrl, string $cancelUrl): Session
    {
        $lineItems = [];
        foreach ($cart as $item) {
            $lineItems[] = [
                'price_data' => [
                    'currency' => 'eur',
                    'product_data' => ['name' => $item['name']],
                    'unit_amount' => (int)($item['price'] * 100),
                ],
                'quantity' => 1,
            ];
        }

        return Session::create([
            'payment_method_types' => ['card'],
            'line_items' => $lineItems,
            'mode' => 'payment',
            'success_url' => $successUrl,
            'cancel_url' => $cancelUrl,
        ]);
    }
}
```

## Service Stripe (StripeService.php)

# Sécurité de l'application



- CSRF Protection : Middleware sur tous les formulaires
- Mots de passe : Hashés avec bcrypt
- Validation : 8 caractères, majuscule, minuscule, chiffre
- Activation : Token aléatoire (32 bytes)
- Rôles : Admin / Client avec hiérarchie

config/packages/security.yaml

```
security:
    password_hashers:
        Symfony\Component\Security\Core\User\PasswordAuthenticatedUserInterface:
            'auto'
    providers:
        app_user_provider:
            entity:
                class: App\Entity\User
                property: email
    firewalls:
        main:
            lazy: true
            provider: app_user_provider
            form_login:
                login_path: app_login
                check_path: app_login
                enable_csrf: true
            logout:
                path: app_logout
    access_control:
        - { path: ^/admin, roles: ROLE_ADMIN }
        - { path: ^/profile, roles: ROLE_USER }
```

# Configuration de sécurité (security.yaml)

# Tests unitaires (PHPUnit)

Tests réalisés :

- RegistrationControllerTest : 2 tests
- SecurityControllerTest : 1 test
- PaymentControllerTest : 2 tests
- UserRepositoryTest : 1 test

Résultat : OK (6 tests, 12 assertions)

Commande : vendor/bin/phpunit



tests/Repository/UserRepositoryTest.php

```
class UserRepositoryTest extends TestCase
{
    public function testFindActiveUsers(): void
    {
        $user = new User();
        $user->setEmail('test@example.com');
        $user->setIsActive(true);

        $this->assertTrue($user->isActive());
        $this->assertEquals('test@example.com', $user->getEmail());
    }

    public function testGetRoles(): void
    {
        $user = new User();
        $this->assertContains('ROLE_CLIENT', $user->getRoles());
    }
}
```

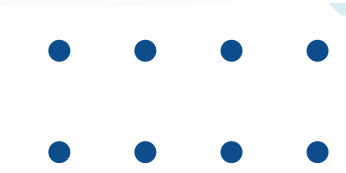
# Tests unitaires (PHPUnit)

# Démonstration

# Recherches effectuées

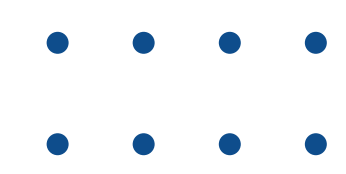


- Symfony 8.1 (framework)
- Doctrine ORM (base de données)
- Twig (templates)
- Stripe API (paiements)
- Mailtrap (email sandbox)
- PHPUnit (tests unitaires)
- Methodologie Merise (conception DB)



# Conclusion

Le projet Knowledge Learning répond à l'ensemble des fonctionnalités demandées :

- ✓ Inscription et activation par email
  - ✓ Authentification sécurisée (admin/client)
  - ✓ Achat de cursus et de leçons (Stripe)
  - ✓ Validation des leçons
  - ✓ Certifications automatiques
  - ✓ Interface responsive
  - ✓ Tests unitaires (6 tests, 12 assertions)
  - ✓ Code commenté en anglais
  - ✓ Documentation complète
  - ✓ Code disponible sur GitHub
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# THANK YOU

Presented by : ESSOUFI RIM

